TOPS TECH Task-7 (CS)

Task-7 Scanning network

1. **Install Nmap on your system and use it to scan your local Wi-Fi network for all connected devices. List the IP addresses and device names (if available) found in the scan.**

   Ans:

   Scan Local Wi-Fi Network Using Nmap

   Objective:
   - The objective of this task is to identify all devices connected to the local Wi-Fi network using the Nmap network scanning tool and list their IP addresses and device names (if available).

   Command Used

   → A network scan was performed using the Nmap command:
   - nmap -sn 192.168.31.0/24

   - nmap: Network scanning tool.
   - -sn: Performs a ping scan without port scanning.
   - 192.168.31.0/24: Scans all devices in the local network range.

   Scan Results

Sr. No.	**IP Address**	**Device Name**	**Status**
1	192.168.31.1	jiofiber	Host Up
2	192.168.31.22	V2250.lan	Host Up
3	192.168.31.28	JAY-AMBE.lan	Host Up
4	192.168.31.152	Kartikey.lan	Host Up

- ➢ The scan successfully detected all active devices connected to the local Wi-Fi network.
- ➢ The router was identified as the gateway device.
- ➢ Multiple client devices such as a desktop computer, smartphone, and laptop were found.
- ➢ Nmap displayed the IP addresses and hostnames of available devices.

⇨ Nmap was successfully used to scan the local Wi-Fi network and identify connected devices. The scan provided information about active hosts, including their IP addresses and device names, helping in understanding the network topology and connected systems.

2. **Install Use the 'arp -a' command in your terminal or command prompt to display the current ARP table. Identify at least two devices on your network and write down their IP and MAC addresses.**

Ans:

Display and Analyze the ARP Table

Objective:

- o The objective of this task is to use the ARP (Address Resolution Protocol) table to identify devices connected to the local network and record their IP and MAC addresses.

Command:

→ arp -a

- ➢ arp: Displays and modifies the Address Resolution Protocol cache.
- ➢ -a: Shows the current ARP table entries for all network interfaces.

ARP Results:

IP Address	**MAC Address**	**Type**
192.168.31.1	aa-bb-cc-dd-ee-ff	Dynamic
192.168.31.22	11-22-33-44-55-66	Dynamic
192.168.31.28	22-33-44-55-66-77	Dynamic

Identified Devices:

Device 1:

→ IP Address: 192.168.31.1
→ MAC Address: aa-bb-cc-dd-ee-ff

Device 2:

→ IP Address: 192.168.1.22
→ MAC Address: 11-22-33-44-55-66

Observations

- The ARP table contains mappings between IP addresses and MAC addresses of devices on the local network.
- Dynamic entries are automatically learned by the system through network communication.
- Each connected device has a unique MAC address associated with its IP address.
- The router is commonly identified by the default gateway IP address.

⇨ The ARP table was successfully displayed using the arp -a command. The IP and MAC addresses of devices on the local network were identified and recorded. This information helps in understanding network communication and device identification within the local network.

3. **Perform a port scan on your own computer using Nmap to find which common ports (like 22, 80, 443) are open. Summarize your findings in a table with port number, service, and status (open/closed).**

Ans:

Perform a Port Scan on the Local Computer Using Nmap

Objective:

- The objective of this task is to perform a port scan on the local computer using Nmap and identify common open ports, the services running on them, and their status.

Command 1:

→ nmap localhost

Command 2:

→ nmap 127.0.0.1

- Nmap is a network scanning tool used to discover hosts and services on a network.
- localhost refers to the current computer being used.
- The scan helps identify open ports and the services associated with them.

Scan Results:

Port Number	**Service**	**Status**
22	SSH	Closed
80	HTTP	Closed
443	HTTPS	Closed
135	MSRPC	Open
445	Microsoft-DS (SMB)	Open
2020	XinUPageserver	Open
2021	ServExec	Open
2022	Down	Open

Port Analysis

Port 22 (SSH):
- → Service: Secure Shell (SSH)
- → Status: Closed
- → Purpose: Used for secure remote login and administration.

Port 80 (HTTP):
- → Service: HyperText Transfer Protocol (HTTP)
- → Status: Closed
- → Purpose: Used for hosting web services and websites.

Port 443 (HTTPS):
- → Service: HyperText Transfer Protocol Secure (HTTPS)
- → Status: Closed
- → Purpose: Provides encrypted and secure web communication.

- ⇨ Nmap scan was successfully performed on the local computer.
- ⇨ Open ports found: 135, 445, 2020, 2021, and 2022.
- ⇨ Closed ports found: 22 (SSH), 80 (HTTP), and 443 (HTTPS).
- ⇨ Ports 135 and 445 are related to Windows networking services.
- ⇨ Open ports indicate that the corresponding services are actively listening for network connections.
- ⇨ Nmap helped identify active network services and assess the system's network exposure.

4. **Simulate a Zomato-style scenario: Imagine you are a security tester for a food delivery app. Use Nmap to check if the test server at 'scanme.nmap.org' has any open ports that could be vulnerable. List any ports found and suggest what kind of services might be running on them.

Hint: Use the command 'nmap scanme.nmap.org' and research the common services for the ports you find.**

Ans:

Scan a Public Test Server Using Nmap

Objective:

- The objective of this task is to scan a public test server using Nmap, identify open ports, and determine the services running on those ports.

Command:

→ nmap -sV scanme.nmap.org

- **nmap**: Network scanning tool used to discover hosts and services.
- **-sV**: Enables service version detection.
- **scanme.nmap.org**: Official public test server provided by the Nmap project for educational and testing purposes.

Scan Results:

Port Number	**Service**	**Status**
22	SSH	Open
80	HTTP	Open
9929	Nping Echo	Open
31337	Elite	Open

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Service Analysis:

Port 22:
- → Service: Secure Shell (SSH)
- → Status: Open
- → Purpose: Provides secure remote access and system administration capabilities.

Port 80:
- → Service: HyperText Transfer Protocol (HTTP)
- → Status: Open
- → Purpose: Hosts web pages and allows users to access websites through a web browser.

Port 9929:
- → Service: Nping Echo Service
- → Status: Open
- → Purpose: Used by Nmap tools for network testing and packet analysis.

Port 31337:
- → Service: Elite
- → Status: Open
- → Purpose: Test service available on the Nmap scan server for educational and scanning purposes.

Observations:
- The Nmap scan successfully detected multiple open ports on the public test server.
- Service version detection identified the services associated with each port.
- Port 22 provides secure remote administration access.
- Port 80 allows web-based communication using HTTP.
- Ports 9929 and 31337 are primarily available for testing and demonstration purposes on the Nmap test server.

5. **Refactor your Nmap scan command to only show open ports and their services, hiding all closed ports from the output.

Constraint: Use the '-open' or '-sV' flag in your command to filter results.**

Ans:

Display Only Open Ports and Services Using Nmap

Objective:

- The objective of this task is to use Nmap to display only the open ports on a target system and identify the services running on those ports.

Command:

- nmap --open -sV scanme.nmap.org

→ **nmap** : Network scanning tool used to discover hosts and services.
→ **--open** : Displays only open ports and hides closed or filtered ports.
→ **-sV** : Detects the service name and version running on each open port.
→ **scanme.nmap.org** : Official public test server provided by the Nmap project for learning and testing purposes.

Scan Results:

Port Number	**Service**	**Status**
22	SSH	Open
80	HTTP	Open
9929	Nping Echo	Open
31337	Elite	Open

Service Analysis:

Port 22:
- → Service: Secure Shell (SSH)
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Port 80:
- → Service: HyperText Transfer Protocol (HTTP)
- → Status: Open
- → Purpose: Hosts web pages and allows users to access websites through a web browser.

Port 9929:
- → Service: Nping Echo Service
- → Status: Open
- → Purpose: Used by Nmap tools for network testing and packet analysis.

Port 31337:
- → Service: Elite
- → Status: Open
- → Purpose: Test service available on the Nmap scan server for educational and scanning purposes.

Observations:
- o The --open option filtered the results and displayed only open ports.
- o Service version detection provided additional information about active services.
- o The scan identified four active services running on the target server.
- o Open ports indicate that the corresponding services are accepting network connections.
- o The results help administrators understand which services are exposed to the network.

- The Nmap scan was successfully performed using the --open and -sV options.
- Only open ports were displayed, making the output easier to analyze.
- The scan identified active services such as SSH, HTTP, Nping Echo, and Elite on the target server.
- This task demonstrates how Nmap can efficiently identify exposed services and assist in network security assessment.